MILITARY SPECIFICATION

WARHEAD, 2.75 INCH ROCKET, PRACTICE: XM274

This specification is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers the quality assurance provisions and special requirements not covered by the drawings for parts and assembly of the XM274 Signature Practice Warhead.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-48077	Primer, M104, Parts for Loading, Assembling, and Packing
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-D-50868	Detonator M85

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331	Environmental Testing and Performance of Fuzes and Components

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC: 1340

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MIL-STD-810	Environmental Test Methods
MIL-STD-1167	Ammunition Data Cards

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

MISSILE INTERIM SPECIFICATION

MIS-30862	Rocket, 2.75 Inch, Complete Round with MK66 Motor: Assembling, Packing, and Testing
MIS-50263	Rocket, 2.75 Inch, Complete Round with MK40 and MK4 Motor: Assembling, Packing, and Testing

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGE DRAWINGS

9370251	Warhead, 2.75 Inch Rocket, Practice: XM274
8886810	Box, Packing, Ammunition, for Warhead, 2.75 Inch Rocket, H.E., M151, Smoke WP M156
8886811	Container, Ammo Fiber, for Warhead, 2.75 Inch Rocket, H.E., M151, Smoke WP M156

INSPECTION EQUIPMENT DRAWINGS

9201212	Envelope Drawing, Safety, and Arming Inspection Equipment

PUBLICATIONS

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, parts 100-199

(The code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS.

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9370251 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 M423 Safe & arm.

3.3.1 Spring embrittlement. None of the springs used in the fabrication of the S&A shall break when extended or distorted to the point of minimum permanent distortion.

3.3.2 Spring set. The setback weight spring shall return within .015 inch of the original free length when compressed and heat treated.

3.3.3 Arming and detent functioning. The rotor assembly in the loaded Safety and Arming Device (S&A) shall rotate from the fully safe position to the fully armed position and the detent shall lock the rotor in the fully armed position.

3.3.4 Rotor reset capability. The rotor assembly shall be capable of returning to the unarmed position on its own after the S&A device has been partially armed.

3.3.5 Arming time. The rotor assembly in the loaded S&A shall rotate from the fully safe position to the fully armed position in not less than 0.67 sec. and not more than 0.88 sec. when subjected to arming acceleration of 27.00 plus or minus 0.25 g's. The S&A shall meet this requirement at any temperature between -50°F ± 5°F and +160°F ± 5°F.

3.3.6 Non-arming test. The rotor assembly in the loaded S&A shall not arm when subjected to acceleration of 11.25 g's minimum for a minimum of three (3) seconds. This test shall be conducted subsequent to the Arming Time Test.

3.3.7 Detonators. All detonators, prior to assembly, shall have been subjected to and passed all requirements of MIL-D-50868.

3.3.8 Primers. All primers, prior to assembly, shall have been subjected to and passed all requirements of MIL-P-48077.

3.4 Interfaces. The warhead shall be compatible with the MK66 Mod 1 and MK40 Mod 3.

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3.5 Fluoroscopic or radiographic inspection. The warhead shall be fluoroscopically or radiographically examined to verify to the extent possible the presence and positioning of all components as well as the absence of foreign material. Warheads selected for first article testing shall be radiographically examined with Government approved equipment prior to static or field firing. Radiographs shall be of a quality which will permit proper evaluation as determined by the responsible technical agency of the Government.

3.6 Warhead weight. The weight shall be 9.3 ± 0.5 pounds.

3.7 Rough handling. The warhead shall withstand rough handling when tested as specified in 4.5.7 and, with the exception of the five foot drop test, shall be ballistically tested as specified in 4.5.12. The warhead shall be safe to handle and dispose of followng the five foot drop test.

3.8 Temperature. The warhead shall withstand exposure to a storage temperature of -50°F ± 5°F, 70°F ± 15°F, and 160°F ± 5°F and shall comply with the ballistic test as specified in 4.5.12.

3.9 Secured cargo vibration. The warhead shall withstand exposure to secured cargo vibration as specified in 4.5.8 and shall comply with the ballistic test as specified in 4.5.12.

3.10 Helicopter vibration. The warhead shall withstand exposure to helicopter vibration as specified in 4.5.9 and shall comply with the ballistic test as specified in 4.5.12.

3.11 Forty-foot drop. The warhead assembled to the rocket motor and packaged for shipment shall withstand a 40 foot drop as specified in 4.5.10 and be safe to dispose of.

3.12 Sealing. The smoke cartridge assembly shall withstand a vacuum of 6.0 ± 1.0 inches of mercury for a minimum period of 60 seconds without leakage, when tested in accordance with 4.5.11.

3.13 Warhead performance. The completely assembled round in flight configuration shall be ballistically tested to comply with the requirements of this specification.

3.14 Data requirements. The contractor shall generate data in accordance with the requirements of the data item descriptions cited in 4.5.13.

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3.15 First article inspection. This specification makes provision for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.16 Process specification - molded or cast parts. When a multiple cavity mold is used in the fabrication of plastic or die cast components, a minimum of five (5) components from each mold cavity shall be included in the first article sample. The components shall bear a mark which identifies the components with the individual cavity.

3.17 Workmanship. Unless otherwise specified on the drawing, all parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, porosity, warpage, burn marks, checks, blisters, excess flash, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing base material. The loading and assembly operations shall be performed in a thorough workmanlike manner consistent with the best practices of the industry. All required markings shall be neat and legible.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are made a part of this specification.

4.2 Classification of inspections. Unless otherwise specified herein or in the contract, the following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Office for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawing	Quantity
Pin, Firing	9370238	15
End Cap, S&A Housing	9370239	15 or 5/cavity whichever is greater
Cap, Nose	9370240	15
Cap, End	9370241	15 or 5/cavity whichever is greater
Disc, Barrier	9370242	25
Pad, Felt	9370243	15
Retainer	9370244	15
Plug	9370245	15
Housing, Smoke and S&A	9370246	15 or 5/cavity whichever is greater
Body, Warhead	9370249	15
Device, Safety and Arming	9215619-1	25
Device, Safety and Arming (Detonator only)	9215617-1	10
Device, Safety and Arming	9215617-1	75
Rotor Housing Assembly	8883719	10
Spring, Detent	8883713	10
Spring, Setback Weight	8883718	10
Gear and Pinion Assy No. 1	8883703	25
Gear and Pinion Assy No. 2	8883700	25
Escapement Gear and Pinion Assembly	8883710	25
Lever Assembly	8883706	30
Rotor Assembly	8883744	20
Inner Plate Assembly	9260804	20
Escapement and Gear Assembly	8883692	20
Smoke Cartridge Assembly	9370252	30
Warhead, XM274	9370251	96

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078 and Table I specified herein.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 6		DRAWING NUMBER
	Warhead, 2.75 Inch Rocket, Practice XM274				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Pin, Firing (Dwg. 9370238) Examination for Defects	15		3.2	4.4.2.1
	End Cap, S&A Housing (Dwg. 9370239) Examination for Defects	15(a)		3.2/ 3.16	4.4.2.2
	Cap, Nose (Dwg. 9370240) Examination for Defects	15		3.2	4.4.2.3
	Cap, End (Dwg. 9370241) Examination for Defects	15(a)		3.2/ 3.16	4.4.2.4
	Disc, Barrier (Dwg. 9370242) Examination for Defects	25		3.2	4.4.2.5
	Pad, Felt (Dwg. 9320243) Examination for Defects	15		3.2	4.4.2.6

NOTES:

(a) Minimum of 5/cavity for multiple mold with at least 15 total.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 6		DRAWING NUMBER
	Warhead, 2.75 Inch Rocket, Practice XM274				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Retainer (Dwg. 9370244) Examination for Defects	15		3.2	4.4.2.7
	Plug (Dwg. 9370245) Examination for Defects	15		3.2	4.4.2.8
	Housing, Smoke and S&A (Dwg. 9370246) Examination for Defects	15(a)		3.2/ 3.16	4.4.2.9
	Body, Warhead (Dwg. 9370249) Examination for Defects	15		3.2	4.4.2.10
	Device, Safety and Arming (Dwg. 9215619-1) Examination for Defects	25		3.2	4.4.2.11
	Device, Safety and Arming (Detonator only) (Dwg. 9215617-1) Detonator Push Test	10		3.2	4.4.2.12

NOTES:

(a) Minimum of 5/cavity for multiple mold with at least 15 total.

AMSMC Form 1570, 1 Feb 85

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 3 OF 6		DRAWING NUMBER
	Warhead, 2.75 Inch Rocket, Practice XM274				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Device, Safety and Arming (Dwg. 9216717-1)				
	Examination for Defects	75		3.2	4.4.2.12
	Arming Time	75(b)		3.3.5	4.5.5.19
	Arming and Detent Functioning	75		3.3.3	4.5.5.16
	Setback Weight	75		3.2	4.5.5.17
	Rotor Reset Capability	75		3.2	4.5.5.18
	Non-Arming	75		3.3.6	4.5.5.20
	Rotor Housing Assembly (Dwg. 8883719)				
	Examination for Defects	10		3.2	4.4.2.13
	Load Test of Spring Stud	10		3.2	4.5.5.12
	Spring, Detent (Dwg. 8883713)				
	Examination for Defects	10		3.2	4.4.2.14

NOTES:

(b) 25 at -50°F ± 5°F; 25 at +70°F ± 5°F; 25 at +160°F ± 5°F. 2 hours minimum temperature conditioning time. Test shall be completed within one minute after removal from temperature conditioning box. First Article shall be rejected if one or more fail minimum time requirement or 2 or more fail maximum time requirement.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 4 OF 6		DRAWING NUMBER
	Warhead, 2.75 Inch Rocket, Practice XM274				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Spring Embrittlement	10		3.3.1	4.5.5.1
	Spring, Setback Weight (Dwg. 8883718) Examination for Defects	20		3.2	4.4.2.15
	Spring Embrittlement	10		3.3.1	4.5.5.1
	Spring Set	10		3.3.2	4.5.5.2
	Gear and Pinion Assembly No. 1 (Dwg. 8883703) Examination for Defects	25		3.2	4.4.2.16
	Load Test of No. 1 Gear and Pinion Assembly	25		3.2	4.5.5.3
	Gear and Pinion Assembly No. 2 (Dwg. 8883700) Examination for Defects	25		3.2	4.4.2.17
	Load Test of No. 2 Gear and Pinion Assembly	25		3.2	4.5.5.4
	Escapement Gear and Pinion Assembly (Dwg. 8883710) Examination for Defects	25		3.2	4.4.2.18

NOTES:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 5 OF 6		DRAWING NUMBER
	Warhead, 2.75 Inch Rocket, Practice XM274				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Load Test of Escapement Gear and Pinion Assembly	25		3.2	4.5.5.5
	Lever Assembly (Dwg. 8883706)				
	Examination for Defects	30		3.2	4.4.2.19
	Load Test of Lever Shaft	10		3.2	4.5.5.6
	Load Test of Lever Pallet	10		3.2	4.5.5.7
	Torque Test of Lever Shaft	10		3.2	4.5.5.10
	Rotor Assembly (Dwg. 8883744)				
	Examination for Defects	20		3.2	4.4.2.20
	Load Test of Annular Gear	10		3.2	4.5.5.8
	Load Test of Lock Roller Pin	10		3.2	4.5.5.9
	Inner Plate Assembly (Dwg. 9260804)				
	Examination for Defects	20		3.2	4.4.2.21
	Load Test of Bridge rivet	20		3.2	4.5.5.11
	Escapement and Gear Assembly (Dwg. 8883692)				
	Examination for Defects	20		3.2	4.4.2.22

NOTES:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 6 OF 6		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
	Warhead, 2.75 Inch Rocket, Practice XM274				
CATEGORY	**EXAMINATION OR TEST**	**NO. OF SAMPLE UNITS**	**AQL OR 100%**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE /INSPECTION METHOD**
	Load Test of Pillars	20		3.2	4.5.5.13
	Smoke Cartridge Assembly (Dwg. 9370252)				
	Examination for Defects	30		3.2	4.4.2.23
	Leakage Test	30(a)		3.12	4.5.11
	Warhead, 2.75 Inch Rocket, Practice: XM274 (Dwg. 9370251)				
	Examination for Defects	96		3.2	4.4.2.24
	Radiographic	96		3.5	4.5.6
	Weight	96		3.6	Scale
	Rough Handling Test	36		3.7	4.5.7
	Secured Cargo Vibration	36		3.9	4.5.8
	Helicopter Vibration	19		3.10	4.5.9
	40 Foot Drop Test	5		3.11	4.5.10
	Ballistic Test (c)	85(b)		3.13	4.5.12

NOTES:

(a) The first article leakage sample shall be rejected if one (1) or more units exhibit leakage.

(b) Sample is composed of items from rough handling (30), secured cargo (36), and helicopter vibration (19).

(c) The first article ballistic sample shall be rejected if two (2) or more units fail to function. Ballistic results of rough handling units shall be excluded.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of XM274 Warheads shall contain:

a. Detonators from one interfix lot number from not more than three lots from one manufacturer.

b. Primers from one interfix lot number from not more than three lots from one manufacturer.

c. M423 S&A bearing the same metal parts lot interfix number from one metal parts manufacturer.

d. Metal parts from no more than two lots bearing the same interfix number from one manufacturer.

The quantity of warheads in each production shall not exceed 20,000.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9370238
4.4.2.1	Pin, Firing				NEXT HIGHER ASSEMBLY 9370251
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR					
101	Diameter		0.40%	3.2	Gage
102	Straightness		0.40%	3.2	Gage
103	Length		0.40%	3.2	Gage
104	Profile of Point		0.40%	3.2	Gage
MINOR					
201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9370239
4.4.2.2	End Cap, S&A Housing				NEXT HIGHER ASSEMBLY 9370252
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR					
101	Diameter at shoulder		0.40%	3.2	Gage
102	Firing pin hole diameter		0.40%	3.2	Gage
103	Mold inspection	5/ cavity		3.2	4.5.5
104	Thickness at firing pin hole		0.40%	3.2	Gage
105	Depth of firing pin from shoulder		0.40%	3.2	Gage
MINOR					
201	Shoulder length		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.17	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9370240
4.4.2.3	Cap, Nose				NEXT HIGHER ASSEMBLY 9370251
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR					
101	Inside diameter		0.40%	3.2	Gage
102	Inside depth		0.40%	3.2	Gage
MINOR					
201	Total length		0.65%	3.2	Gage
202	Wall thickness		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9370241
4.4.2.4	Cap, End				NEXT HIGHER ASSEMBLY 9370252
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR					
101	Shoulder diameter		0.40%	3.2	Gage
102	Mold inspection	5/ cavity		3.2	4.5.5
MINOR					
201	Shoulder length		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.17	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.5	Disc, Barrier		1 OF 1		9370242
					NEXT HIGHER ASSEMBLY 9370252
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR					
101	Diameter		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
MINOR					
201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9370243
4.4.2.6	Pad, Felt				NEXT HIGHER ASSEMBLY 9370252
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR	None Defined				
MINOR					
201	Thickness		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9370244
4.4.2.7	Retainer				NEXT HIGHER ASSEMBLY 9370251
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR					
101	Cartridge interface diameter		0.40%	3.2	Gage
MINOR					
201	Outside width (3 places)		0.65%	3.2	Gage
202	Cartridge interface radius (12 places)		0.65%	3.2	Gage
203	Thickness		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9370245
4.4.2.8	Plug				NEXT HIGHER ASSEMBLY 9370251
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR 101	Diameter		0.40%	3.2	Gage
MINOR 201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9370246
4.4.2.9	Housing, Smoke and S&A				NEXT HIGHER ASSEMBLY 9370252
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR					
101	Depth at S&A end		0.40%	3.2	Gage
102	Diameter at S&A end		0.40%	3.2	Gage
103	Diameter at smoke end		0.40%	3.2	Gage
104	Web thickness		0.40%	3.2	Gage
105	Mold inspection	5/ cavity		3.2	4.5.5
MINOR					
201	Outside diameter at S&A end		0.65%	3.2	Gage
202	Outside diameter at smoke end		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9370249
4.4.2.10	Body, Warhead				NEXT HIGHER ASSEMBLY 9370251
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR					
101	Cartridge interface diameter		0.40%	3.2	Gage
102	Nose cap interface diameter		0.40%	3.2	Gage
103	Groove width		0.40%	3.2	Gage
104	Bourrelet diameter		0.40%	3.2	Gage
105	Pitch diameter of threads		0.40%	3.2	Gage
106	Inside lip width		0.40%	3.2	Gage
107	Vent counterbore diameter		0.40%	3.2	Gage
108	Perpendicularity		0.40%	3.2	Gage
109	Intrusion length		0.40%	3.2	Gage
110	Intrusion diameter		0.40%	3.2	Gage
MINOR					
201	Wrench interface diameter		0.65%	3.2	Gage
202	Smallest inside diameter		0.65%	3.2	Gage
203	Groove diameter		0.65%	3.2	Gage
204	Inside diameter at base end		0.65%	3.2	Gage
205	Large diameter at nose end		0.65%	3.2	Gage
206	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9215619-1
4.4.2.11	Device, Safety and Arming				NEXT HIGHER ASSEMBLY 9215617-1
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL 1	Distance between annular gear and No. 2 gear, max.		100%	3.2	Gage
MAJOR 101	Alignment of detonator cavity with the outside diameter across the rotor housing posts		0.40%	3.2	Gage
102	Torque test or rotor stop stud		0.40%	3.2	4.5.5.14
MINOR 201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9215617-1
4.4.2.12	Device, Safety and Arming				NEXT HIGHER ASSEMBLY 9270252
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Setback weight return capability		100%	3.2	4.5.5.17
2	Rotor reset capability		100%	3.2	4.5.5.18
3	Non-arming		100%	3.3.6	4.5.5.20
MAJOR					
101	Height from forward face to base of top of rotor, max.		0.40%	3.2	Gage
102	Distance from rotor flat edge of base, min.		0.40%	3.2	Gage
103	Primer inverted (red disc should be observed through the top)		0.40%	3.2	Visual
104	Primer not securely staked		0.40%	3.2	Visual
105	Primer perforated or deformed by staking		0.40%	3.2	Visual
106	Detonator inverted (yellow disc should be observed through aft end)		0.40%	3.2	Visual
107	Detonator perforated or damaged by staking		0.40%	3.2	Visual
108	Foreign matter present		0.40%	3.2	Visual
109	Lubricant missing		0.40%	3.2	Visual
110	Diameter across posts of rotor housing, min.		0.40%	3.2	Gage
111	Protective finish missing or incomplete		0.40%	3.2	Visual
112	Markings missing or incomplete		0.40%	3.2	Visual

NOTES:

(a) Primers and detonators are to be tested in accordance with Para 4.5.2 and 4.5.3 respectively.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.12	Device, Safety and Arming		2 OF 2		9215617-1
					NEXT HIGHER ASSEMBLY 9270252
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
113	Distance from detonator disc to bottom of S&A rotor housing (armed position)		0.40%	3.2	Gage
114	Detonator push test (detonator only)		0.40%	3.2	4.5.5.15
115	Arming and detent functioning		100%	3.3.3	4.5.5.16
116	Arming time		100%(b)	3.3.5	4.5.5.19
MINOR 201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:
(b) 100% at 70°F ± 5°F; 25 at -50°F ± 5°F; 25 at +160°F ± 5°F. Two hours minimum temperature conditioning time. Testing shall be completed within one minute after removal from temperature conditioning box. The lot shall be rejected if one or more fail minimum time requirement or two or more fail maximum time requirement.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883719
4.4.2.13	Rotor Housing Assembly				NEXT HIGHER ASSEMBLY 8883686
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR					
101	Load test of spring stud		0.40%	3.2	4.5.5.12
102	Rotor housing arm width		100%	3.2	Gage
MINOR					
201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883713
4.4.2.14	Spring, Detent				NEXT HIGHER ASSEMBLY 8883692
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR 101	Spring embrittlement	10		3.3.1	4.5.5.1
MINOR 201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883718
4.4.2.15	Spring, Setback Weight				NEXT HIGHER ASSEMBLY 9215619
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR					
101	Spring embrittlement	10		3.3.1	4.5.5.1
102	Spring set	10		3.3.2	4.5.5.2
MINOR					
201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883703
4.4.2.16	Gear and Pinion Assembly No. 1				NEXT HIGHER ASSEMBLY 8883692
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Load test of No. 1 gear and pinion assembly		100%	3.2	4.5.5.2
2	Maximum gap between gear and pinion		100%	3.2	Gage
MAJOR	None defined				
MINOR					
201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.17	Gear and Pinion Assembly No. 2		1 OF 1		8883700
					NEXT HIGHER ASSEMBLY 8883692
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Load test of No. 2 gear and pinion assembly		100%	3.2	4.5.5.4
2	Maximum gap between gear and pinion		100%	3.2	Gage
MAJOR	None defined				
MINOR					
201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883710
4.4.2.18	Escapement Gear and Pinion Assembly				NEXT HIGHER ASSEMBLY 8883692
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Load test of escapement gear and pinion assembly		100%	3.2	4.5.5.5
2	Maximum gap between gear and pinion		100%	3.2	Gage
MAJOR	None defined				
MINOR					
201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883706
4.4.2.19	Lever Assembly				NEXT HIGHER ASSEMBLY 8883692
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Load test of lever shaft		100%	3.2	4.5.5.6
2	Load test of lever pallet		100%	3.2	4.5.5.7
3	Maximum gap between lever and shaft		100%	3.2	Gage
MAJOR					
101	Torque test of lever shaft		0.40%	3.2	4.5.5.10
MINOR					
201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883744
4.4.2.20	Rotor Assembly				NEXT HIGHER ASSEMBLY 9215619
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL 1	Load test of annular gear		100%	3.2	4.5.5.8
MAJOR 101	Load test of lock roller pin		0.40%	3.2	4.5.5.9
MINOR 201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9260804
4.4.2.21	Inner Plate Assembly				NEXT HIGHER ASSEMBLY 8883692
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR 101	Load test of bridge rivet		0.40%	3.2	4.5.5.11
MINOR 201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883692
4.4.2.22	Escapement and Gear Assembly				NEXT HIGHER ASSEMBLY 9215619
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR 101	Load test of pillars		0.40%	3.2	4.5.5.13
MINOR 201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9270252
4.4.2.23	Smoke Cartridge Assembly				NEXT HIGHER ASSEMBLY 9370251
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	M423 S&A device in unarmed position		100%	3.2	Visual
MAJOR					
101	Sealing test		100%	3.12	4.5.11
102	Outside diameter after sealing, max.		0.40%	3.2	Gage
MINOR					
201	Weight of smoke composition		0.65%	3.2	Scale
202	Marking incorrect		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-70467 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Warhead, 2.75 Inch Rocket, Practice: XM274	1 OF 1	9370251
			NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Marking and identification band incorrect		100%	3.2	Visual
102	Fluoroscope		100%	3.5	4.5.6
103	Ballistic Test	36		3.13	4.5.12
MINOR					
201	Evidence of poor workmanship		0.65%	3.17	Visual

NOTES:

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4.4.3 Testing. Testing is described in the Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein. The contractor shall submit all inspection equipment designs to the Government for evaluation and approval prior to fabrication of such equipment.

4.4.5 Contractor inspection system. As part of contractor's approved quality procedure, there shall be a provision for checking the adequacy of inspections made in connection with listed critical or major defects. These procedures shall include the following provisions which are applicable to the manufacture and inspection of S&A's.

4.4.5.1 Use of simulated defect standards. The following listed verified inspections shall be validated and inspection equipment verified hourly using Design Agency approved item facsimilies or standards containing the defects and supplied by the contractor. The contractor shall resupply the item facsimilies when it is determined by the Government that the facsimilies are no longer representative enough of the product to make it unidentifiable to the contractor inspection personnel, or it is not functioning in the necessary manner and cannot be corrected satisfactorily. These standards should be representative of defects actually found or anticipated. They should be inert and readily identifiable to the Government to ensure recovery should they be inadvertently introduced into production. Simulated defect standards used to verify the following verified inspection should be constructed and identified in such a manner that they are not readily recognizable to the contractor inspection personnel as defect standards. Standards should be used so that their number, time, and sequence of introduction into the inspection cycle is known only to those personnel responsible for checking the accuracy of the operation. Adequacy of the standards shall be verified at least once each shift.

Verified Inspection

Paragraph 4.5.5.17 - Setback weight return capability.

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NOTE: When there is automatic detection and automatic rejection of defective items to a controlled area or a controlled rejection of defective items, it is permissible for the defect standards to be identifiable and their introduction into production noticeable to contractor inspection personnel.

4.4.5.2 Undetected defect standards. Provisions shall be established for the corrective action to be taken where the equipment or contractor inspection personnel fail to identify a standard as defective. These provisions shall include a reinspection of all acceptable material subsequent to the last acceptable verification check as a minimum.

4.5 Methods of inspection.

PRECAUTION: This specification covers sampling and testing of chemical, toxic, or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations, and procedures must be followed in handling and processing these materials.

4.5.1 Materials, components, and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspection. The visual, manual, and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Check test for deterioration of primers. If the total time between original acceptance of any primer lot and the assembly of that lot into the S&A's exceeds two years, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass the functioning and output tests specified in MIL-P-48077 immediately before the primer lot is assembled into the S&A's. This test shall be performed by the contractor on primers selected by the Government Inspector at the facility assembling the primers into the rotors. The test shall be conductd at Government expense without cost to the contractor who loaded the primers or the contractor who assembled the primers into the S&A's. The test shall

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not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer was under the jurisdiction of either contractor.

4.5.3 Check test for deterioration of detonators. If the total time between original acceptance of any detonator lot and the assembly of that lot into the S&A's exceeds two years, at any time since previous tests, the detonator lot shall be subjected to and must satisfactorily pass the functioning and output tests specified in MIL-D-50868 immediately before the detonator lot is assembled into the S&A's. The test shall be performed by the contractor on detonators selected by the Government Inspector at the facility assembling the detonators into the rotors. The test shall be conducted at Government expense without cost to the contractor who loaded the detonators or the contractor who is to assemble the detonators into the S&A's. The test shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the detonator lot was under the jurisdiction of either contractor.

4.5.5 Dimensional control of molded or cast parts. In place of normal sampling and inspection associated with the Classification of Defects, a sample of five (5) parts as molded from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraphs 4.4.2.2, 4.4.2.4, and 4.4.2.9 after continuous production of each 5000 parts or at the end of each week, whichever occurs first. Of the sample, one (1) part shall be the last produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each type of defect according to MIL-STD-105, using the AQL's detailed in the Classification of Defects. All cavities removed from production because of some fault may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

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4.5.5 S&A tests.

4.5.5.1 Spring embrittlement. The spring shall be stretched along its longitudinal axis in the case of a coil spring, or bent in the case of a flat spring, until the spring no longer returns to its original position. When the spring no longer returns to its original length or profile, it shall be considered distorted permanently. Any spring that breaks prior to this point shall be classed defective.

4.5.5.2 Setback weight spring set. The springs shall be measured at their original free length, assembled in a restraining fixture in accordance with 4.4.4 and compressed to a normal height of .750 inches, placed in an oven and subjected to a temperature of not more than 180 degrees Fahrenheit (^{O}F) or less than 170°F for 24 hours. The springs shall be removed from the oven and be cooled to ambient temperature. After cooling the springs for 24 hours minimum, the springs shall be measured and the measurement shall be compared to the original free length to ascertain compliance with the specified requirement. Any spring that fails the requirement shall be classed defective.

4.5.5.3 Load test of No. 1 gear and pinion assembly. Apply an axial load prpendicular to the No. 1 gear and in such a direction as to test the assembly of the No. 1 gear and pinion. The required force shall act against the No. 1 gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any No. 1 gear and pinion assembly that fails the requirement shall be classed defective.

4.5.5.4 Load test of No. 2 gear and pinion assembly. Apply an axial load prpendicular to the No. 2 gear and in such a direction as to test the assembly of the No. 2 gear and pinion. The required force shall act against the No. 2 gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any No. 2 gear and pinion assembly that fails the requirement shall be classed defective.

4.5.5.5 Load test of escapement gear and pinion assembly. Apply an axial load perpendicular to the escapement gear and in such a direction as to test the assembly of the escapement gear and pinion. The required force shall act against the escapement gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any escapement gear and pinion assembly that fails the requirement shall be classed defective.

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4.5.5.6 Load test of lever shaft. Apply the axial load perpendicular to the lever and in such a direction as to test the assembly of the lever and lever shaft. The required force shall act against the lever. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any lever shaft that fails the requirement shall be classed defective.

4.5.5.7 Load test of lever pallet. Apply an axial load perpendicular to the lever and in such a direction as to test the assembly of the lever and the lever pallet. The required force shall act against the lever. Upon removal of the required force, the height of the lever pallet above the lever shall be inspected for compliance with the drawing requirement. Any lever pallet that fails the requirement shall be classed defective.

4.5.5.8 Load test of annular gear. Apply a load perpendicular to the rotor and in such a direction as to test the assembly of the annular gear studs to the annular gear and the rotor. The required force shall act against the rotor. Upon removal of the required force, the distance from the face of the rotor to the outside face of the annular gear shall be inspected for compliance with the drawing requirement. Any annular gear that fails the requirement shall be classed defective.

4.5.5.9 Load test of lock roller pin. Apply an axial load perpendicular to the rotor and in such a direction as to test the strength of the assembly of the lock roller pin to the rotor. The required force shall act against the rotor. Any lock roller pin that fails the requirement shall be classed defective.

4.5.5.10 Torque test of lever shaft. A torque shall be applied between the lever and the lever shaft. Any lever shaft that fails the requirement shall be classed defective.

4.5.5.11 Load test of bridge rivet. Apply the required load perpendicular to the inner plate and into the bridge rivet from the side opposite the bridge. The required force shall act against the inner plate. Any bridge rivet that fails the requirement shall be classed defective.

4.5.5.12 Load test of spring stud. Apply an axial load perpendicular to the rotor housing and in such a direction as to test the assembly of the spring stud and the rotor housing. The required force shall act against the rotor housing. Any spring stud that fails the requirement shall be classed defective.

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4.5.5.13 Load test of pillars. Apply the required axial load perpendicular to the inner place and in such a direction as to test the assembly of the pillars to the inner plate and outer plate. The required force shall act against the inner plate. Any pillars that fail the requirement shall be classed defective.

4.5.5.14 Torque test of rotor stop stud (eccentric pin). The required torque shall be applied between the rotor stop stud and gear assembly. The required force shall act on the gear assembly. Any rotor stop stud that fails the requirement shall be classed defective.

4.5.5.15 Load test of detonator. Apply a five pound minimum axial load perpendicular to the rotor and in such a direction as to test the strength of the staking of the detonator in the rotor. Any staked detonator that fails the requirement shall be classed defective.

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

4.5.5.16 Arming and detent functioning. The S&A, in the safe position, shall be assembled to the test equipment in accordance with 4.4.4. The equipment shall accelerate the Safety and Arming device to 20.75 g's maximum for a minimum of three plus one, minus zero seconds in a direction which simulates rocket launch. At the completion of acceleration, the equipment shall be halted and visual observation shall be made for the locking position of the detent. Finger tip pressure shall be applied to the rotor to rotate in the safe direction to ascertain positive arming. At the completion of the test, the safety and Arming Device shall be returned to the safe position. Any S&A that fails the requirement shall be classed defective.

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

4.5.5.17 Setback weight return capability. This test shall be performed as specified on drawing 9215617. S&A's failing to comply with the specified requirements shall be classed defective and removed from the lot. Defect standards for verification of the inspection process shall be in accordance with 4.4.4.3 and 4.4.4.4. Defect standards for use with automatic equipment shall be as follows:

a. Units with setback weights which bind when depressed and remain in the depressed condition.

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b. Units with setback weights which start to return but do not completely return to the original locking position.

c. Units with setback weights which return from the depressed condition at a slower than normal rate causing rejection.

4.5.5.18 Rotor reset capability. The loaded S&A shall be assembled to the test fixture in accordance with 4.4.4. The rotor shall be rotated to a position at least 26 degrees, but not more than 35 degrees, from the fully safe position and released. Observation shall be made for rotor return to the fully safe position. S&A's failing to comply with the specified requirement shall be classed defective and removed from the lot.

4.5.5.19 M423 S&A arming time test. The loaded S&A with the g-weight in the downward position shall be assembled to the test equipment in accordance with 4.4.4. The equipment shall accelerate the Safety and Arming Device in a clockwise direction to 27.00 g's plus or minus 0.25 g's and the arming time shall be recorded to the nearest hundredth of a second using a calibrated timer. S&A's failing to comply with the specified requirements shall be classed defective and removed from the lot.

4.5.5.20 M423 S&A non-arming test. The loaded S&A with the g-weight in the downward position shall be assembled to the test equipment in accordance with 4.4.4. The equipment shall accelerate the Safety and Arming Device in a clockwise direction to 11.25 g's minimum for a minimum of three (3) seconds. Observation shall be made for rotor in the safe position. S&A's failing to comply with the specified requirements shall be classed defective and removed from the lot.

4.5.6 Radiographic. X-ray and fluoroscopically view each round in two planes, 90° opposed. Verify that all major components, assemblies, and sub-assemblies are present and properly placed, that the S&A in the unarmed position and no foreign material is present. Any warhead that fails the requirements shall be removed from the lot.

4.5.7 Rough handling test. The rough handling test shall be performed as follows on each of thirty-six (36) warheads for first article testing only:

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Each warhead shall be packed and packaged in accordance with MIS-30862 or MIS-50263, whichever is applicable, except that dummy 2.75 Inch Rocket Motors shall be substituted. The test warheads are to be positioned in the box with the warhead nose in one direction and a dummy load utilized in the remaining fiber container. The dummy load shall be located in the upper right position as viewed from the nose end. The warheads shall be equally divided and temperature conditioned to -50°F ± 5°F, 70°F ± 15°F, and 160°F ± 5°F. The conditioning time shall be 8 hours minimum and 24 hours maximum.

Each box shall be dropped seven feet onto a steel plate which is solidly supported by concrete. The orientations of the boxes on striking are:

a. Eighteen (18) warheads (6 boxes--2 from each temperature condition) warhead nose down (longitudinal axis 45° from vertical).

b. Eighteen (18) warheads (6 boxes--2 from each temperature condition) warhead base down (longitudinal axis 45° from vertical).

No individual box shall be subjected to more than one drop. The drop equipment and stand shall be capable of providing an unimpeded free fall prior to striking the plate and permit rebounding. Prior to performing this drop test, the following is necessary:

a. Serialize each warhead.

b. X-ray or fluoroscope each round in two planes, 90° opposed. Verify all components are secure and the S&A is in the unarmed position.

c. Before dropping, each packaged warhead shall be visually examined for evidence of incorrect packaging.

After performing the drop test on all warheads, each round shall be x-rayed or fluoroscoped. A visual examination shall be performed. The first article shall be rejected if one or more warheads are found to be not safe as a result of the effects of the seven-foot drop test. If no safety hazard is noted, the warheads will be removed from the boxes and placed on an approved package tester in their fiber containers for the loose cargo test. The warheads shall not be tied down in any manner inside the tester. The warheads shall be equally divided and temperature conditioned to -50°F ± 5°F, 70°F ± 15°F, and 160°F ± 5°F. The conditioning time shall be 8 hours minimum and 24 hours maximum. The tester shall then be vibrated for 30 minutes at a frequency of 5.0 hertz at 1.3 g's.

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Following the loose cargo test, the warheads shall be subjected to visual and x-ray or fluoroscopic examination. The first article shall be rejected if one or more warheads are found to be not safe as a result of the effects of the loose cargo test.

If no safety hazard is noted, thirty (30) warheads shall be ballistically tested in accordance with 4.5.12 for information only. The remaining six (6) warheads (two conditioned to each specified temperature) shall be subjected to an unpackaged five-foot drop test onto a steel plate which is solidly supported by concrete. The drop orientations of the warheads (at each specified temperature) on striking will be:

a. One warhead nose down with longitudinal axis 45 degrees from vertical.

b. One warhead nose down with longitudinal axis vertical.

Following the five-foot drop test, the warheads shall be safe to handle or dispose of. The first article shall be rejected if one or more warheads are found to be not safe as a result of the effects of the five-foot drop test. Warheads subjected to the five-foot drop test shall not be ballistically tested.

4.5.8 Secured cargo vibration test. The secured cargo vibration test shall be performed as follows on each of thirty-six (36) warheads for first article testing only.

The sample warheads shall be packed and packaged in accordance with MIS-30862 or MIS-50263, whichever is applicable, except that dummy 2.75 Inch Rocket Motors shall be substituted. The test warheads are to be positioned in the box with the warhead nose in one direction and a dummy load utilized in the remaining fiber container. The dummy load shall be located in the upper right position as viewed from the nose end. The warheads shall be equally divided and temperature conditioned to -50°F ± 5°F, 70°F ± 15°F, and 160°F ± 5°F. The conditioning time shall be 8 hours minimum and 24 hours maximum.

The packed shipping boxes shall be subjected to the vibration test in accordance with Method 514.2, Equipment Category g, Procedure x, Table 514.2-VII, and Figure 514.2-7 Curve AW of MIL-STD-810C. Resonance testing does not apply.

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Prior to performing this test, the following is necessary:

a. Serialize each round.

b. X-ray or fluoroscope each round in two planes 90° apart. Verify all components are secure and the S&A is in the unarmed position.

Following the vibration test, each round shall be x-rayed or fluoroscoped. A visual examination shall be performed. The first article shall be rejected if one or more warheads are found to be not safe as a result of the effects of the secured cargo vibration test.

If no safety hazard is noted, the warheads shall be ballistically tested in accordance with 4.5.12.

4.5.9 Helicopter vibration test. The helicopter vibration test shall be performed as follows on each of nineteen (19) warheads for first article testing only.

a. Sweep test. Warheads with inert motors shall be loaded into a light weight launcher and subjected to a low level sweep test. The loaded launchers will be tested in each of three orthogonal axes. The sweep level will be 1/2 G or lower as necessary to limit the maximum response of the launchers to the Table II levels. This test will define the dynamic resonances of the system, as mounted on a vibration shaker.

b. Dwell test. Following the sweep tests, warheads with inert motors loaded in a launcher shall be subjected to vibration excitation in each of the three orthogonal axes. The test shall be conducted at each of four principal frequencies 11, 22, 33, and 44 Hz. The input levels at each frequency will be such as to provide the launcher response levels as specified in Table III for the time durations indicated. The test method to be used is the random noise method limited to the frequency range of interest.

c. Prior to performing the helicopter vibration test, x-ray or fluoroscope each round in two planes, 90° opposed, to verify that all components are secure and the S&A is in the unarmed position. After performing the test on all warheads, each round shall be x-rayed or fluoroscoped. A visual examination shall be performed. The first article shall be rejected if one or more warheads are found to be not safe as a result of the effects of the helicopter vibration test. If no safety hazard is noted, the warheads shall be ballistically tested in accordance with 4.5.12.

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TABLE II. Sweep test

Launcher Condition	Direction of Excitation		
	Transverse (G)	Vertical (G)	Axial (G)
FULL	4.3	5.8	3.1

NOTE: The sweep time shall be 1 octave per minute 5 to 100 to 5 Hz.

4.5.10 Forty foot drop test. Five warheads shall be subjected to the forty foot drop test as specified in MIL-STD-331, Test 103.1, Procedure 1 (substitute warhead for fuze) for first article testing only. The warheads shall be packed and packaged in accordance with 5.1 except that dummy 2.75 Inch Rocket Motors shall be substituted. One test round shall be packaged with three dummy rounds in each box. The first article shall be rejected if one or more warheads function or are found to be unsafe for disposal as a result of the effects of the forty foot drop test.

4.5.11 Sealing test. The smoke cartridge assembly shall be tested in a water tank equipped for observation under vacuum. The water temperature shall be 65°F minimum to 75°F maximum. The water tank shall be equipped so that the smoke cartridge assembly must be raised above the water level while still under vacuum before releasing the vacuum and removing the smoke cartridge assembly.

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TABLE III. Dwell test.

2.75 INCH ROCKET VIBRATION TEST LEVELS (MINIMUM)

CONDITION	TIME PER AXIS (HOURS)	LOC	VERTICAL AXIS					TRANSVERSE AXIS					LONGITUDINAL AXIS				
			11 HZ G^2/HZ	22 HZ G^2/HZ	33 HZ G^2/HZ	44 HZ G^2/HZ	TOT GRMS	11 HZ G^2/HZ	22 HZ G^2/HZ	33 HZ G^2/HZ	44 HZ G^2/HZ	TOT GRMS	11 HZ G^2/HZ	22 HZ G^2/HZ	33 HZ G^2/HZ	44 HZ G^2/HZ	TOT GRMS
19 TUBE LOADED	.8*	FWD	.017	.05	.03	.004	.38	.015	.01	.026	.0015	.29	.03	.005	.005	.005	.3
		AFT	.05	.02	.016	.05	.6	.2	.015	.008	.004	.58	.025	.002	.0003	.002	.22

*For the loaded condition only, 4 minutes of the .8 hours per axis shall be at a level of 2 X the total GRMS.

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The uppermost surface of the smoke cartridge assembly, when in a horizontal position, shall be submerged 4 to 6 inches below the water surface and subjected to a vacuum of 6.0 ± 1.0 inches of mercury for a minimum period of 60 seconds. Leaks are indicated by air bubbles issuing from the smoke cartridge assembly. Do not mistake the escape of occluded air for leakage. Defectives are smoke cartridge assemblies failing to meet the requirements of 3.12.

4.5.12 Ballistic test. The requirement for the warhead is to provide a visible signature at impact with a functional reliability of greater than 93% when fired with MK66 or MK50 motors at ranges from 1000 meters to 6000 meters. Performance shall be evaluated by ground launching a sample of 36 XM274 warheads (18 assembled with MK66 Mod 1 motors and 18 assembles with MK40 Mod 3 motors). 12 rockets, (6 of each type) shall be conditioned at -50°F ± 5°F, 12 rockets (6 of each type) at 70°F ± 15°F, and 12 rockets (6 of each type) at 160°F ± 5°F for minimum of eight hours. Two of each type at each temperature shall be fired from an M260 or M261 launcher to impact at 1000 meters, two of each type at each temperature shall be fired to impact at 3000 meters, two of each type at each temperature shall be fired to impact at 6000 meters. If three or more warheads fail to function, the lot shall be rejected.

4.5.13 Data requirements. The contractor shall submit a quality inspection test demonstration and evaluation report in accordance with Data Item Descriptions DI-R-1721 and DI-R-1724 giving the detailed test results.

5. PACKAGING.

5.1 Packaging and marking. When shipping as complete rounds, MIS-30862 or MIS-50263 as applicable applied. When shipping as warheads only, drawings 8886810 and 8886811 and CFR49, Parts 100.199 apply.

5.2 Ammunition data cards. Data card information shall be as specified in MIL-STD-1167.

6. NOTES.

6.1 Intended use. The warheads covered by this specification are intended for use as part of the 2.75 Inch Hydra 70 Rocket System.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See 6.2 of MIL-A-48078. Submit designs as required to Commander, US Army Armament Research, Development and Engineering Center (ARDEC), ATTN: AMSMC-QAF-I(D), Dover, NJ 07801-5001.

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6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph	Data Requirement	Applicable DID
4.6.13	Quality Inspection Defect Report	DI-R-1721
4.6.13	Quality Inspection Test Demonstration and Evaluation Report	DI-R-1724

(Data item descriptions related to this specification, and identified in Section 6, will be approved and listed as such in DoD 6000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.5 First article. When a first article is required for inspection and approval, see 3.15 and 4.3.

6.6 Government furnished parts and support. The procuring agency will provide support components and facilities for ballistic testing.

6.7 Source control and specification control drawings. Certificates of conformance from the supplier certifying that the item complies with the drawing requirements shall be provided to the Government representative.

Custodian:
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Preparing activity:
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